

AquaWatch Naija: Hyperlocal Water Quality Monitor

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Project Overview

Throughout the world, millions of citizens still have inadequate access to safe drinking water. To combat this growing issue, the United Nations (UN) Sustainable Development Goal 6 (SDG 6) was created to “ensure access to water and sanitation for all” (United Nations, n.d.). AquaWatch Naija was developed by Team Quantum Engineers for the Grow with Google: Mentor Me Collective (MMC) BUILD project to support this goal.

As defined in the problem statement provided by the MMC, Nigeria is a country where “community members lack a central, accessible public dashboard to view municipal water testing results and safety warnings.” Nigeria Health Watch reports that, according to 2024 data released by the Nigerian Institute of Water Engineers (NIWE), “approximately 179 million Nigerians lack access to safely managed drinking water” (Nigeria Health Watch, n.d.). Therefore, water data transparency is critical. Unsafe water, sanitation, and hygiene services (WASH) can contribute to increased mortality rates in Nigerian neighborhoods, causing the most vulnerable residents to be exposed to waterborne illnesses such as diarrheal, cholera, and other diseases. Having a comprehensive dashboard provides residents with a needed resource that helps them make the most informed safety decisions.

Research & Data Strategy

The data infrastructure behind this water quality monitor portal was built using a Kaggle dataset (Yadav, 2024) combined with two academic case studies that were selected to represent a hyperlocal, Nigeria-specific water quality overview. The Kaggle dataset explores water pollution and disease data across multiple countries. For this project, the data was filtered down to Nigeria-only records, covering the South, Central, East, North, and West regions. This collection of data contained 25 years of water data from 2000-2024, and the main testing results used in the project included water source type, pH, turbidity, contaminant and bacterial levels, treatment method, and disease incidence rates.

Relevant metrics from the case studies were extracted and standardized into a format that matched the Kaggle dataset columns. The Jabi Lake case study (Adejuwon et al., 2025), which sampled the lake in 2021, assessed the surface water quality during the wet and dry seasons. It provided the Abuja Federal Capital Territory (FCT) data. The Benue South¹ case study (Edegbene et al., 2025), based on 2023 fieldwork, contributed data for the Benue South Senatorial District. It examined the water quality of sources across three Local Government Areas (LGAs) within Benue State, Nigeria: Apa, Ohimini, and Otukpo. Both case studies shared common numerical test results from the Kaggle dataset, specifically pH and turbidity (NTU) measurements. Disease Indicator / Risk was a column category added to the combined dataset, and it is used to provide a more accurate snapshot of these areas.

AquaWatch Naija Interprets the Data & Provides the Solution

AquaWatch Naija uses the combined dataset to power the dashboard, which is deployed on Streamlit Community Cloud with Python as the codebase and GitHub as the central repository. The interface is a user-friendly, web portal app. CSS was used to style the dashboard with a glassmorphic design that’s inviting and easy to use.

When navigating to the homepage, a Location Filter on the left-hand side allows users to select the Region and Water Source Type. AquaWatch Naija uses a Pandas- and NumPy-based pipeline for cleaning, aggregation, and standardization. This pipeline isolates key contamination thresholds and safety indicators from the combined dataset and visualizes them via banners such as Total Samples, Safe Sources, and Active Alerts. The dashboard’s safety methodology measures specific factors for the safety alerts. The pH threshold range of 6.5–8.5 reflects World Health Organization (WHO) drinking water guidance on operational parameters.² The Turbidity (≤ 5.0 NTU) threshold is informed by WHO’s broader acceptability guidance on turbidity. WHO notes that turbidity becomes visible to consumers at around 4 NTU and recommends that treatment systems target levels well below 1 NTU. Bacteria (≤ 1000 CFU/mL) and Contaminant Level (≤ 7.0 ppm) are not WHO figures. These are dataset-calibrated cutoffs used because the source data doesn’t identify which pathogen or contaminant was tested.³ Taking these factors into

¹ Benue South is not the same as the broad “South” region from the Kaggle dataset, which spans multiple, unrelated parts of the country. “South” and “Benue South” are two distinct regions from two different sources and are kept separate throughout the dataset and the dashboard.

² WHO does not set a formal, health-based guideline value for pH.

³ Unlike these dataset-calibrated cutoffs, WHO’s actual standard for bacteria like *E. coli* is zero detectable presence in any 100 mL sample.

account, a site is High Risk if two or more readings breach a threshold, Moderate Risk if exactly one does, and Safe otherwise. For the nine Abuja/Benue South rows missing the numeric Bacteria/Contaminant Level readings, a qualitative field note (e.g., “Extreme” or “High”) acts as an independent contamination signal. Those sites are screened using field observations, along with pH and turbidity, to determine the final risk level, which reflects a more severe evaluation. Throughout the dashboard, there’s a safety alert guide that uses a stoplight system where red is High Risk, orange is Moderate Risk, and green is Safe. It is incorporated throughout the dashboard.

On the dashboard, an interactive, color-coded Water Safety Map uses pydeck to highlight testing sites, with points corresponding to the safety alert colors. The map’s hover tooltip pulls information from the tables for the Year, Region, Water Source, Risk Level, and Reason.

Above the Water Safety Map, there are three tabs: Water Safety Map, Water Quality Breakdown, and Community Action Steps. On the main tab, the Water Safety Status by Region table incorporates shared water quality metrics that fuel the safety classification methodology. In the tables and map, all results are tagged by year. The Water Quality Breakdown tab provides a more detailed analysis, including a Municipal Water Testing Results table with additional columns, such as Risk Level and Disease Indicator / Risk. The two columns are separate checkpoints. Risk Level compares water samples against established contamination thresholds. Disease Indicator / Risk evaluates the cases of Cholera, Typhoid, or Diarrheal disease (per 100,000 people) and populates the cell with the appropriate record. For Abuja (FCT) and Benue South, the original contamination and Disease Indicator / Risk field notes are used to calculate the Risk Level, and those notes are retained in the Disease Indicator / Risk cells.

The last tab, Community Action Steps, directs residents on ways to interpret and act on their local results. It explains how to check the Disease Indicator / Risk column in the Municipal Water Testing Results table and what those readings indicate. In addition, residents can find practical advice on treating water in the table’s Water Treatment Method column. Users are also provided instructions on when to report and request additional testing. Altogether, AquaWatch Naija gives residents access to vital water testing results, as well as a dashboard that municipal agencies can use to store data efficiently and inform the public in real time. It lays the groundwork for future capabilities that can further support the UN SDG 6.

Implementation Plan

The BUILD project window for Team Quantum Engineers ran from July 15, 2026, to August 14, 2026. To meet the project’s requirements, the team divided the workload into five phases over four weeks. Beginning with phase one (week 1), the team researched Nigeria’s water quality reporting and discovered the Kaggle dataset. The selected data provided the information needed for the water quality monitor and was filtered to Nigeria-only data. Then, relevant data from two peer-reviewed case studies that aligned with the key findings in the Kaggle data were extracted, standardized, and merged into a final combined dataset via Google Sheets.

In phase two (week 2), the finalized dataset was reviewed and converted to a CSV file, and the team’s GitHub repository was created. From there, we held multiple working sessions to develop the Python code for the Streamlit dashboard, using the AI assistant Claude to help draft and troubleshoot. To mitigate risk, the code was peer-reviewed, and every AI-assisted change was further tested and adjusted by the team before deployment. We also created a requirements.txt file that pins exact package versions (e.g., streamlit==1.60.0) rather than open-ended version ranges. This protects the deployed app from styling changes and new version releases unless we deliberately upgrade and retest.

Once we were satisfied with the finalized codebase, we entered phase three (week 3). The dashboard was deployed to the Streamlit Community Cloud, with the team’s GitHub repository serving as the controlled repository for the code. The Streamlit link to the web app was tested and reviewed for errors. It was also run locally, and the team confirmed the setup instructions worked. From there, we drafted the README and written project summary.

Next, we recorded the project walkthrough video, edited it, and linked it to the README in phase four (week 4). Throughout this phase, we continued to review all project materials (the dataset, codebase, dashboard, README, video, and written summary) and made the necessary updates.

To conclude phase five (week 4), we conducted a final review of all AquaWatch Naija materials. We submitted the finished project via a pull request from the team-quantum-engineers branch to the MMC 2026-cohort branch.

References

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